Molly Pate

EN.605.617

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Module 9 Write Up

**Goal of Assignment:**

The primary goal of this project was to implement a GPU of Compressed Sensing for MRI images, specifically targeting the "Scan Time" problem, as described here: https://pmc.ncbi.nlm.nih.gov/articles/PMC4984938/. As described in the article, the core issue in clinical imaging isn't a compute time limitation, but rather an MRI machine hardware constraint. Traditional scans require 20 minutes of full k-space sampling. This means if a patient moves during that window, the entire frequency dataset is corrupted, resulting in a blurry, unusable image.

On a very simple level, I wanted to simulate a solution to this physical bottleneck by using the GPU to process incomplete MRI-like data. To model this, I utilized the cuFFT library to move an image into the k-space, where MRI data is naturally represented. I then used the cuSPARSE library to apply a digital under sampling mask, treating the sampled frequencies as a sparse matrix in CSR format.

While this implementation does not perform full compressed sensing reconstruction, it demonstrates the core idea of under sampling by intentionally discarding a large portion of the frequency data and reconstructing an approximate image. By doing this, I wanted to complete a simulation with CUDA libraries, but also perhaps lay the groundwork for more advanced reconstruction techniques that could recover missing information.

**Challenges:**

One of the main challenges in this project was first understanding what I was actually trying to simulate from an MRI perspective. The concept of transforming an image into k-space using a FFT was not immediately intuitive, and I had to spend time learning how pixel data becomes a set of frequency coefficients that represent the image in a completely different domain. Once I understood that, a second challenge was figuring out what to do with the sparse data after undersampling. Mathematically, this meant realizing that I was multiplying the k-space vector by a diagonal matrix that selectively keeps or removes frequency components. Translating that idea into code using cuSPARSE required understanding how CSR format encodes a sparse matrix and how the cusparseSpMV operation applies that matrix to a vector. Ensuring that the indexing and structure of the sparse matrix correctly aligned with the flattened k-space data was not straightforward, and it required me to connect the abstract linear algebra concepts directly to how data is stored and processed on the GPU. Overall, I did appreciate though that the libraries in this assignment really abstracted away most of the underlying math/efficiency issues, so that I could actually focus on the high level steps at hand.

**Triumphs:**

I had never worked with images and GPUs together before, and in this project I was able to take an image, transform it into k-space using cuFFT, apply a sparse sampling mask with cuSPARSE, and reconstruct a visible image, despite the scale of what I was working with probably being trivial for the GPU. I also gained a clearer understanding of how abstract linear algebra, like how multiplying by a sparse diagonal matrix maps to real GPU computations. On top of that, it was just interesting to see what an MRI machine does under the hood, even beyond the assignment. 😊